

ICT ACADEMY – CYBER SECURITY PROGRAM

Day 3 – Cloud, Linux & Security Fundamentals with AWS EC2 Practical Lab

Learning Log / GitHub Documentation

Session Details

Participant	Mohammad Jawad
Day	Day 3
Date	28 September 2026
Hands-on Platform	Amazon Web Services (AWS)
Cloud Service Used	Amazon EC2
Operating System	Ubuntu 26.04 LTS

Day 3 Focus

Build on cloud fundamentals by learning about AWS Regions and Availability Zones, practice core Linux commands, and study Identity and Access Management (IAM/IGA) alongside the CIA triad. Apply these concepts hands-on by connecting to an AWS EC2 Ubuntu server, running Linux commands, and cloning a GitHub repository directly from the cloud instance.

1. AWS Regions & Availability Zones

- Each AWS region contains a minimum of 3 Availability Zones (AZs), and a maximum of 6.
- Availability Zones are nothing but isolated data centers within a region.

2. Linux Commands (Day 03)

Command	Purpose
pwd	Print working directory.
last	Shows who the last logged-in users were.
w	Identify how many users are logged in on the current system.

To Install "last" (if missing)

```
sudo apt update -y
sudo apt upgrade -y
sudo apt install last
```

3. Linux Basic Commands

1) **Navigation** – move between directories.

Linux Terminal Cursor Movement Shortcuts	
Shortcut	Action
←	Move left 1 character
→	Move right 1 character
Ctrl + A	Beginning of line
Ctrl + E	End of line
Alt + B	Back one word
Alt + F	Forward one word
Ctrl + ←	Back one word
Ctrl + →	Forward one word
Home	Beginning of line
End	End of line

4. CIA Triad		
Letter	Term	Meaning
C	Confidentiality	Prevent unauthorized disclosure.
I	Integrity	Prevent unauthorized modification.
A	Availability	Ensure authorized access when needed.

5. IAM (Identity Access Management)
The 7 A's of IAM – "IAAAAAAA"

IAM Components	
Letter	Term
I	Identity
A	Authentication
A	Authorization
A	Auditing
A	Accessing
A	Accounting
A	Administration
A	Availability

In IAM
- Identity is obtained from Active Directory. Tools: Microsoft Entra ID, Okta, AWS IAM, Google Cloud IAM.

6. IGA (Identity Governance Administration)
- The remaining 7 A's (beyond Identity) are controlled by IGA. Tools: SailPoint, Saviynt.

7. Your EC2 CIA Lab (AWS EC2 – Linux)		
C – Confidentiality	I – Integrity	A – Availability
IAM / MFA	SHA256	Backup
SSH	FIM (File Integrity Monitoring)	Monitoring
Permissions	Git Audit	Health Check & Recovery
Encryption		

8. Practical Lab – AWS EC2 (Ubuntu Server)
Goal: Launch an Ubuntu EC2 instance, connect through SSH, run core Linux administration commands, verify network connectivity, and clone a GitHub repository directly from the cloud server.

Lab Steps	
Step	Description
Step 1 – Launch the instance	Launch a new EC2 instance using the Ubuntu 26.04 LTS AMI.
Step 2 – Connect using SSH	From Windows PowerShell, connect using the downloaded private key (.pem) with the command: <code>ssh -i "path-to-key.pem" ubuntu@YOUR_EC2_PUBLIC_IP</code> .
Step 3 – Accept host key	Confirm the ED25519 key fingerprint prompt by typing "yes" to add the host to the list of known hosts.
Step 4 – Verify login	The Ubuntu welcome banner confirms a successful login, showing system load, memory usage and the instance's private IPv4 address.
Step 5 – Update packages	Run <code>sudo apt update -y</code> to refresh the package index from the Ubuntu archive mirrors.
Step 6 – Install required tools	Install supporting packages such as <code>sysvinit-utils</code> , <code>last</code> , <code>iverilog</code> and <code>gtkwave</code> using <code>sudo apt install</code> .
Step 7 – Practice core commands	Run <code>pwd</code> , <code>last</code> and <code>w</code> to explore the working directory and check logged-in users.
Step 8 – Test connectivity	Run <code>ping -c 4 google.com</code> to confirm the instance has working outbound network access.
Step 9 – File operations	Practice basic file handling using <code>wc</code> , <code>grep</code> and <code>rm</code> on a sample text file and test folder.
Step 10 – Clone repository	Run <code>git clone</code> directly on the Ubuntu EC2 instance (not the local Windows shell) to pull a GitHub repository, then authenticate with a GitHub username and token.

Key Troubleshooting Note
Running package installs or git commands directly in local Windows PowerShell failed: <code>sudo</code> is disabled on the local machine, and PowerShell does not support Linux-style redirection syntax such as <code><repository-url></code> . These commands only work correctly when run inside the SSH session on the remote Ubuntu EC2 instance.

SSH Command Used in the Practical
<code>ssh -i "path-to-your-key.pem" ubuntu@YOUR_EC2_PUBLIC_IP</code>
Important: Never upload the .pem private key, passwords, access keys or other secrets to GitHub.

Practical Evidence – AWS EC2 (Ubuntu Server)

1. SSH Connection

```
Windows PowerShell ubuntu@ip-172-31-46-245: ~
PS C:\Users\ DELL> ssh -i "C:\Users\ DELL\Downloads\lad.pem" ubuntu@13.60.189.244
The authenticity of host '13.60.189.244 (13.60.189.244)' can't be established.
ED25519 key fingerprint is SHA256:Sp0u8uHwH7d5ZowY2p2ha2808uYd0p9v18k4fc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.60.189.244' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Mon Sep 28 05:07:20 UTC 2026

System load: 0.03      Temperature: -273.1 C
Usage of /: 30.5% of 6.6GiB    Processes: 121
Memory usage: 26%      Users logged in: 0
Swap usage: 0%          IPv4 address for ens3: 172.31.46.245
```

Connecting via SSH with the .pem key; ED25519 fingerprint accepted; Ubuntu 20.04 LTS welcome banner shown.

2. Shell Ready

```
Windows PowerShell ubuntu@ip-172-31-46-245: ~
Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-46-245:~$
```

System info banner (load, memory, IPv4) followed by a ready shell prompt.

3. Updating Packages

```
ubuntu@ip-172-31-46-245:~$ sudo apt update -y
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4596 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [291 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [680 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/main amd64 Components [212 kB]
Fetched 23.6 MB in 31s (5075 kB/s)
Reading package lists... Done
Building dependency tree... Done
All packages are up to date.
```

sudo apt update -y fetching package indexes from Ubuntu mirrors.

4. Installing Packages

```
Windows PowerShell ubuntu@ip-172-31-46-245:~$ ...
Unpacking libutp0:amd64 (0.75.0-Subuntul) ...
Selecting previously unselected package libpan-utp0:amd64.
Preparing to unpack ../libpan-utp0_0.75.0-Subuntul_amd64.deb ...
Unpacking libpan-utp0:amd64 (0.75.0-Subuntul) ...
Setting up libutp0:amd64 (0.75.0-Subuntul) ...
Setting up libpan-utp0:amd64 (0.75.0-Subuntul) ...
Setting up utp0 (0.75.0-Subuntul) ...
Converting utp login records into utp0 database ... done
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.40-2ubuntu2) ...
Scanning processes... Scanning linux images...

Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-46-245:~$ sudo apt install sysvinit-utils -y
sysvinit-utils is already the newest version (3.15-Subuntul).
sysvinit-utils set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 186
ubuntu@ip-172-31-46-245:~$
```

Package install/config output; kernel up to date, no restarts required.

5. Connectivity & File Ops

```
ubuntu@ip-172-31-46-245:~$ ping -c 4 google.com
PING google.com (142.251.38.110) 56(84) bytes of data:
64 bytes from icarna-ac-in-f14.la100.net (142.251.38.110): icmp_seq=1 ttl=113 time=3.18 ms
64 bytes from icarna-ac-in-f14.la100.net (142.251.38.110): icmp_seq=2 ttl=113 time=3.17 ms
64 bytes from icarna-ac-in-f14.la100.net (142.251.38.110): icmp_seq=3 ttl=113 time=3.16 ms
64 bytes from icarna-ac-in-f14.la100.net (142.251.38.110): icmp_seq=4 ttl=113 time=3.17 ms
--- google.com ping statistics ---
 4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 3.167/3.174/3.182/0.006 ms
ubuntu@ip-172-31-46-245:~$ wc sample.txt
0 0 0 sample.txt
ubuntu@ip-172-31-46-245:~$ grep "text" sample.txt
Building dependency tree... Done
Building dependency tree... Done
All packages are up to date.
ubuntu@ip-172-31-46-245:~$ rm sample.txt test_folder/ -rf
ubuntu@ip-172-31-46-245:~$ exit
logout
Connection to 13.60.189.244 closed.
PS C:\Users\ DELL> sudo apt install lsof
Sudo is disabled on this machine. To enable it, go to the Developer Settings page.
PS C:\Users\ DELL> git clone -repository-url
The < > operator is reserved for future use.
* CategoryInfo          : ParserError: (1) [], ParentContainsErrorRecordException
* FullyQualifiedErrorId : RedirectionNotSupported
PS C:\Users\ DELL>
```

ping -c 4 google.com succeeds; wc/grep/rm used; local PowerShell attempts fail.

6. Cloning the Repository

```
ubuntu@ip-172-31-46-245:~$ ...
Setting up librsync2:amd64 (2.6.1-3dfsg-3) ...
Setting up glycin-loaders (2.1.1-4ds-Subuntul) ...
Setting up glycin-thumbailers (2.1.1-4ds-Subuntul) ...
Setting up libglycin-2.0:amd64 (2.1.1-4ds-Subuntul) ...
Setting up libpdx-pubuf-2.0:amd64 (2.44-3dfsg-Subuntul) ...
Setting up gtk-update-icon-cache (4.22.4-4ds-Subuntul) ...
Setting up adwaita-icon-theme (50.0-1) ...
update-alternatives: using /usr/share/icons/Adwaita/cursor.theme in auto mode
Setting up libgtk-3-0:amd64 (3.24.52-Subuntul) ...
Setting up librsync2-common:amd64 (2.6.1-3dfsg-3) ...
Setting up gthumb (2.3.12-1) ...
Setting up libgtk-3-bin (3.24.52-Subuntul) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
Processing triggers for libpdx-pubuf-2.0:amd64 (2.44-3dfsg-Subuntul) ...
Scanning processes... Scanning linux images...

Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-46-245:~$ git clone https://github.com/akshaya/repository
Cloning into 'repository'...
Username for 'https://github.com':
```

git clone run on the EC2 instance; prompts for GitHub credentials.

9. What I Learned on Day 3

1. Learned that AWS regions contain multiple isolated Availability Zones for redundancy.
2. Practiced essential Linux commands: pwd, last and w.
3. Learned Linux terminal cursor movement shortcuts for faster navigation.
4. Reviewed the CIA triad and its meaning in a security context.
5. Understood the 7 A's of IAM and the tools that implement them.
6. Learned how IGA complements IAM by governing the remaining A's.
7. Mapped CIA triad concepts onto a real EC2 Linux lab environment.
8. Connected to an Ubuntu EC2 instance using SSH and a private key.
9. Updated and installed packages on a live cloud Linux server.
10. Tested network connectivity and practiced basic file-handling commands.
11. Learned the difference between running commands locally versus on a remote SSH session, and cloned a GitHub repository from the EC2 instance.